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Estimating the cascading global impacts of gas disruptions in Qatar

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Abstract

This study examines the global impacts of a localized disruption in Qatar's gas sector using a multi-region input–output framework and scenario-based analysis. While direct impacts of this disruption on importing countries is clear, indirect and cascading impacts are not well understood. We use a Multiregional input-output (MRIO) model to understand the impact of this disruption and whether trade reallocations and increased production can mitigate the effects. Our analysis shows that this disruption leads to significant gas supply losses in Asia and Europe, with the largest aggregate impacts observed in India, China, and South Korea. Allowing for trade reallocation partially mitigates these losses. Further expansion of production capacity among major gas-producing countries enhances supply conditions and generates broader output gains, though these benefits remain concentrated in larger economies. Overall, the results highlight the uneven distribution of both vulnerabilities and recovery potential within global supply chains. While adaptive mechanisms such as trade reallocation and production expansion can alleviate the effects of supply shocks, their effectiveness is limited and heterogeneous. The findings underscore the importance of network structure in shaping shock propagation and resilience, offering insights for managing systemic risks in an interconnected global economy.

Keywords Supply-chain, disruption, cascading impact, multi-regional input-output (MRIO)

Significance statement

This study provides a framework to understand how a localized disruption in Qatar's gas sector can propagate through global supply chains and affect economies globally. By combining multi-region input–output modeling with scenario-based analysis, it identifies key transmission pathways, and the uneven distribution of impacts across countries. The work incorporates adaptive responses, such as trade reallocation and production expansion, to assess the resilience of global systems. The results highlight both the vulnerability of interconnected economies to concentrated supply shocks and the limits of mitigation strategies in achieving supply chain resilience.

Introduction and Background

RM: Research Reports describe the results of original research of exceptional importance. The preferred length of these articles is 6 pages, but PNAS Nexus allows articles up to a maximum of 12 pages. A standard 6-page article is approximately 4,000 words, 50 references, and 4 medium-size graphical elements (i.e., figures and tables).

MM: Figures need better captions, e.g. CHN-XYZ is not clear. Second, is there any recent other papers on this. Cite recent articles in important news sources as well. Anil: can we do a weighted network theoretic analysis.

The analysis does not have time component correct? Can that be considered? You will need to make assumptions on how much time it takes for the dependencies to propagate I suppose.

Qatar plays a pivotal role in the global gas system as one of the world's largest exporters of liquefied natural gas (LNG), with production concentrated in facilities such as Ras Laffan Industrial City. 20% of globally traded LNG originates from Qatar, making international markets highly sensitive to disruptions in its output. Its main hub, Ras Laffan Industrial City, is the largest LNG complex in the world and a key node in global energy supply chains. Major importers include Asia

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(China, India, Japan, South Korea) and parts of Europe, making Qatar central to both regions' energy security.

Additionally, most Qatari exports transit through the Strait of Hormuz, a narrow and geopolitically exposed corridor, which adds to the systemic risk. Thus, any disruption in Qatar's energy production and distribution has global consequences. Recent disruptions to Qatar's gas sector have been driven by a combination of geopolitical tensions and infrastructure damage. Iranian strikes hit key LNG infrastructure (Ras Laffan, Mesaieed) in early March 2026 causing 17% of Qatar's LNG export capacity to go offline, which is about 13 million tons per year, with potential recovery timelines extending over several years (12).

Qatar gas disruptions are having a very significant impact across the world, especially on countries in Asia and Europe. In India and Pakistan, where a large share of LNG and Liquefied Petroleum Gas (LPG) imports come from Qatar and transit through the Strait of Hormuz, supply tightening is already affecting LPG distribution, leaving millions unable to cook and forcing the closure of numerous small businesses and restaurants (2). In the Philippines, rapidly rising fuel prices have led to the declaration of a national emergency and prompted measures such as shortened workweeks, while Pakistan has also closed schools to conserve energy (16).

While some of these countries are importers of gas from Qatar, there are reports of effects of the war even in countries which do not have any direct trade relationships with Qatar. For example, even though the US is a major producer and a net exporter of gas, a disruption in Qatar's gas sector can still affect US households through higher prices. When supply from Qatar falls, global gas prices rise, causing US gas to be diverted to higher paying export markets. This will increase domestic price in the US and raise energy bills for US households. There has been limited study of the cascading effects of disruptions. Aside from Qatar, Australia, Norway, Russia, Canada and Algeria are also major LNG producers. There has been interest in understanding whether these producers can fill supply gaps by reallocation of flows and increased production (though there are many challenges in increasing production, as we discuss later).

Multiregional input-output (MRIO) models are widely used to capture interdependencies across sectors and countries allowing shocks to ripple through global production networks (5). Firm-level and network-based studies show that disruptions can generate significant indirect effects, particularly when supply chains exhibit limited substitutability and complex network structures (10; 19). Linear programming methods have also been used to determine production and final demand in MRIO models. Extensions to energy systems show that interregional trade can mitigate demand deficits following production disruptions, although outcomes depend on network structure and capacity constraints (8).

However, conventional MRIO models rest on fixed production structures that do not allow for reallocation of trade across alternative suppliers under binding constraints, and can yield economically infeasible outcomes under simultaneous supply and demand shocks (15). Koks and Thissen (13) combine multiregional supply-use structures with optimization to assess disaster impacts, incorporating supply constraints, production inefficiencies and allowing for substitution across suppliers with reallocation determined through cost-minimization under capacity constraints. These results suggest that while cascading effects are captured, existing approaches do not represent how production and trade adjust under binding constraints. In these applications, reallocation is determined through optimization rather than by disruptions to existing trade relationships under explicit constraints. As a result, they do not capture how production and trade adjust when reallocation is constrained by existing trade patterns and capacity constraints.

In this study, we develop a global MRIO linear programming model that incorporates disruption, reallocation and capacity expansion using the GTAP 12 database. We use it to examine how shocks to Qatar's gas sector (`QAT-GAS`) propagate through international supply chains. We construct three hypothetical disruption scenarios, including different levels of gas disruptions, partial reallocation by six largest gas-producing countries (`TOP-GAS`) outside the Gulf region, and increased production by them in addition. Together, these scenarios allow us to disentangle the relative importance of supply shocks, trade reallocation, and production responses in impacting global outcomes. Our modeling framework captures both direct effects (reduced gas supply to importing countries) and indirect effects (downstream impacts on industries that rely on gas sector as an input). A reduction in gas supply does not only affect the energy sector; it cascades through the economy, leading to broader output losses. In a global MRIO setting, these effects can be traced across countries, showing how a single, highly connected node, i.e., Qatar's gas sector, can generate ripple effects through other sectors and countries. Furthermore, by comparing outcomes across scenarios, we can assess the extent to which global supply chains exhibit resilience or vulnerability, and identify key regions and sectors that are most exposed to such shocks. This approach provides a systematic basis for understanding the broader economic implications of energy supply disruptions and the potential effectiveness of alternative adjustment mechanisms.

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Results

We consider three disruption scenarios (Table 1): (1) All trade linkages (edges) originating from Qatar's gas sector are reduced uniformly by 30%, 50%, and 70%, representing varying levels of supply contraction. This scenario captures the shock to global gas supply and its cascading effects on dependent sectors and countries, with no mitigation; (2) The six largest gas-producing countries outside the Gulf region partially reallocate their exports to global markets, with adjustments of 10%, 20%, and 30%, reflecting trade diversion and substitution effects under constrained supply conditions; (3) These same producers increase their primary inputs by 5%, 10%, and 20% in the gas sector. To accommodate the additional production, enabled by the increase in primary resources, global trade edge capacities are increased proportionally, enabling the redistribution of supply across the network. We develop a linear programming approach to find solutions to MRIO models for these scenarios using the GTAP 12 database, which covers 163 countries and 65 sectors (see Section 4 for details).

Setting ($\lambda_1 = \lambda_2$)	Parameter	Scenario 1	Scenario 2	Scenario 3
Disruption	Edges	`QAT-GAS -> ALL`	`QAT-GAS -> ALL`	`QAT-GAS -> ALL`
	δ	$\{0.3, 0.5, 0.7\}$	0.3	0.3
Capacity Increase	Edges	-	`TOP-GAS -> ALL`	`GLOBAL`
	α	-	$\{0.1, 0.2, 0.3\}$	$\{0.05, 0.1, 0.2\}$
	Firms	-	-	`TOP-GAS`
	β	-	-	$\{0.05, 0.1, 0.2\}$

Table 1. Scenarios add details about the scenarios, even though there is some repetition (SS: Write what the parameters α, β, δ mean.)

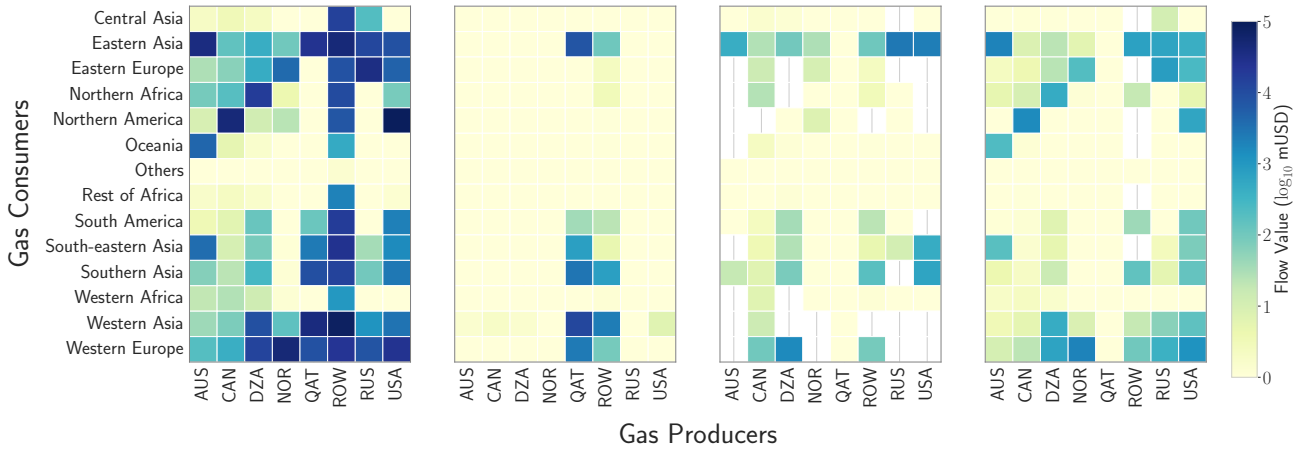


Figure 1 Gas flows from producers to consumer regions. (a) Flows at $(\delta = 0)$. (b) Decline in flows at $(\delta = 0.3)$. (c) Change in flows at $(\delta = 0.3, \alpha = 0.3)$ w.r.t. $(\delta = 0.3)$. (d) Change in flows at $(\delta = 0.3, \alpha = 0.05, \beta = 0.05)$ w.r.t $(\delta = 0.3)$. In fig c, α is 0.1 right, not 0.3? (SS: It's a little unintuitive to have producers as the columns and consumers as rows, since we imagine flows being from producers to consumers. Also, write something about what the insights from the figures are. It is hard to interpret because the meaning of the colors is effectively inverted from (a) to (b). In (a) dark means high flow, while in (b) dark means high reduction in flow. Is there a way to make these more consistent and easier to interpret?)

Scenario 1: Disruption without any mitigation

We model the disruption by proportionally reducing the flow capacities of edges from `QAT-GAS` to all other nodes controlling the disruption percentage with a parameter δ. In this baseline scenario, all other flow capacities remain unchanged, which implies that the flows cannot reallocate to mitigate the demand loss. In Figure 1(a), we show the original gas flows from major gas producers to different regions of the world as a heatmap, illustrating the dependency structure on gas. These gas flows can be used either as intermediate inputs (i.e., as input to other firms) or as final demand by the households. Figure 1(b) reports the loss of gas flows due to the disruption (i.e., the original gas flows minus the flows under disruption) (SS: Your parenthetical comment suggests that the figure is showing the reduced gas flows—original minus disruptions—but the figure looks like it is showing the amount by which flows are reduced, i.e., the "loss of gas flows," as you write before the parenthetical remark.). The impacted regions are largely those which were directly dependent on Qatar gas. While the impact is global, it is most concentrated in Asia, where several regions experience gas supply shortfalls of up to 10 billion USD[1]. Western Europe also incurs losses on the order of billions. Beyond Qatar, gas flows from the rest of the world (ROW), which includes other Gulf producers, are affected to a lesser extent, indicating indirect dependence on Qatari gas as an input into their own production. This secondary disruption further amplifies losses, particularly across Asia and Europe. Just a comment on Figure 1 - x axis shows gas producers and y-axis shows gas consumers. In IO, the usual convention is x-axis shows producers and y-axis shows consumers.

Next, we examine the cascading effects of these gas flow disruptions on the overall economy of countries around the world. Figure 2 shows the losses in final demand and gross output (compared to the original values) aggregated at a country level and visualized as global heatmaps. Nearly all countries are affected to some degree, but regions in Asia and Europe are affected the most, aligning closely with the gas flow losses seen in Figure 1(b).

In terms of loss in final demand, the large Asian economies of India, China, Korea are heavily affected, followed by Italy and the US. This is directly connected to the loss in gas supply from Qatar as seen in Figure 1(b). Notably, the US is an exception here, as direct flow of Qatar gas to the US is negligible. This shows the impact of indirect channels of disruption where the directly affected Asian sectors propagate production losses to their US customers. Output losses are somewhat more contained but have overlapping geographic regions, with additional impacts observed in Taiwan. This suggests that gas from Qatar is a critical input to the industries in these regions, and that a disruption in this input can cascade through their highly integrated internal economies. Figure 3

[1] All USD values reported are in 2023 USD.

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shows the aggregate losses for the top 5 countries as the disruption parameter δ varies. As δ increases from 0.3 to 0.5, the losses nearly double for India and China, whereas losses in other countries rise more gradually. This difference reflects the varying dependence on gas imports and downstream processing capacity across economies.

Scenario 2: Disruption combined with reallocation

Under the same disruption, we now assume selected flows can be reallocated to satisfy the demands unmet by the disruption. We control the degree to which a flow can be reallocated by a parameter α. For a given edge, the flow can increase by up to $\alpha\%$ over its original value, and decrease by up to $\alpha\%$ from its Scenario 1 value [2]. We choose the adaptive edges to be all the outgoing edges from a set of six top gas-producing firms by gross output (excluding QAT-GAS) called TOP-GAS. These are the gas sectors of the US, Russia, Norway, Canada, Australia and Algeria.

Note that these sectors cannot increase their production as they are still limited by their inputs, but can only reallocate out-flows to their recipient firms. Our LP approach (Section 4) finds the most optimal allocation under these constraints giving a lower-bound on the losses possible. Figure 1(c) shows the change in distribution of gas flows to different regions (compared to scenario 1) when reallocation is allowed by TOP-GAS. We find significant reallocation of gas to the affected regions with Asia and Western Europe being the major beneficiaries. While Eastern Asia sees increase in gas flows from all the major gas producers, especially Russia and the US, Western Europe relies more on increased Algerian and Canadian gas supply. The US redirects its gas flows mostly towards Asia instead of Europe. Notably, Western Asia benefits little from the reallocation, with Southern Asia also losing out to East Asia in attracting larger gas flows. This suggests that when given additional flow capacities, gas firms tend to prioritize systemically important sectors in the world, many of which are located in East Asia and Western Europe, if the goal is to maximize the global output and final demand.

When we look at overall economy figures changes in Figure 4, we see the gas flow changes cascade into wider improvements in output and final demand in Asia and Southern Europe. Countries such as China, Korea, Italy translate the gains in gas flows into increased economic output. Figure 5 shows the loss in aggregate final demand and outputs for the top 5 countries as the reallocation parameter α varies while keeping the disruption parameter δ fixed. We observe that the recovery due to this adaptation is not uniform. Countries like the US, Korea, China reduce their losses by up to two orders of magnitude as α is doubled from 0.1 to 0.2. This contrasts with India which sees much more modest improvement, while Pakistan shows no improvement despite the possibility of reallocation.

[2] The limit on the decrease in flow is imposed to ensure that the new solution does not deviate too much from the solution obtained in Scenario 1.

Scenario 3: Disruption mitigated through increased gas production

In this scenario, we assume the TOP-GAS firms increase their primary input by $\beta\%$, to help fulfill the unmet demands due to the disruption. For any given firm, effective utilization of this additional capacity requires corresponding increases in both intermediate inputs and outbound flow capacities. In the absence of sufficient downstream flow capacity, firms may be unable to realize higher production levels without generating excess or unused output.

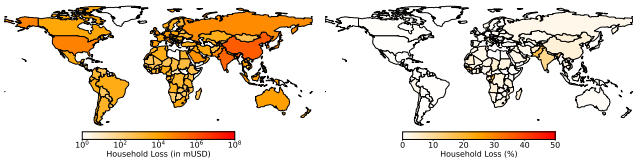
Accordingly, we first augment the primary inputs of the TOP-GAS firms by $\beta\%$. Even a localized increase in demand or production can propagate through the network, inducing higher input requirements across interconnected sectors and regions. This creates a non-trivial allocation problem: which specific edge capacities should be expanded to support the intended production increase? To address this, we adopt a simplified approach: we allow a uniform expansion of flow capacities across all edges, while restricting the increase in primary inputs to the selected TOP-GAS firms. This ensures that the augmented production can be effectively distributed throughout the network without binding capacity constraints. Just like in Scenario 2, we control the reallocation by parameter setting $\alpha = \beta$. Each edge can increase its flow by $\alpha\%$ over its original value and can decrease by up to $\alpha\%$ below its Scenario 1 values.

Figure 1(d) shows the change in distribution of gas flows from major producers to different regions as compared to the Scenario 1 values. With the additional capacity for production, all TOP-GAS countries improve their supply to the affected regions as well as their own economies. Invariably, the domestic gas demand improves greatly for each producer country. The gains for Eastern Europe and Western Asia intensify as compared to Scenario 2 (in Figure 1(c)). Differing from Scenario 2, there is increased Australian gas flow to Southeastern Asia, Russian gas to Eastern and Western Europe, and American gas to South America and Western Europe. Western Asia gas supply is now much improved compared to Scenario 2 bolstered with additional flows from Algeria and the US. There is only modest increase in flows to Southern Asia even though the region faces large shortfalls in gas.

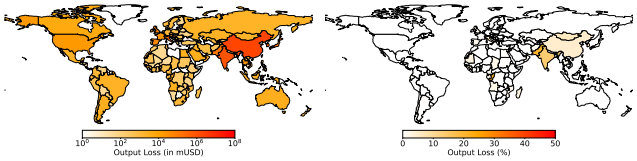
In Figure 6, we plot the global heatmap of changes in final demand flows and gross output per country. We measure these changes with respect to Scenario 1 values. The combination of production expansion and increase in global flow capacity substantially mitigates losses, particularly in large economies such as the US, China, and India. At the same time, reallocation redistributes impacts on final demand, with countries in the Americas experiencing relatively greater losses compared to Scenario 1. Although final demand flows remain negative overall, gains in gross output are observed across much of the world, including heavily affected economies such as China, India, and Italy.

When isolating the effect of production increases among the TOP-GAS group, the benefits accrue disproportionately to countries with strong integration into global gas-dependent supply chains. In particular, economies such as India, China, and South Korea capture a significant share of the gains, alongside the TOP-GAS countries themselves. Figure 7 shows the sensitivity of the losses of the 5 most affected countries to the parameter

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(a) Final Demand Loss under ($\delta = 0.3$)



(b) Output Loss under ($\delta = 0.3$)

Figure 2 Heatmap of global losses under Scenario 1, in absolute values and in percentages

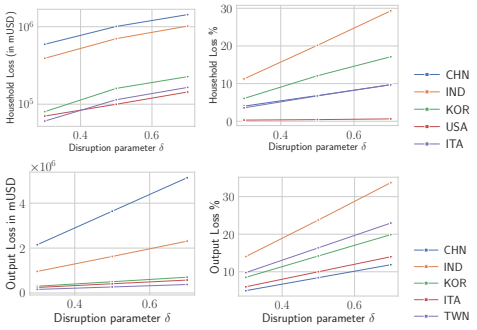


Figure 3 Top 5 most impacted countries in terms of loss with respect to different disruption parameter δ



(a) Final Demand change w.r.t. ($\delta = 0.3$)



(b) Output change w.r.t. ($\delta = 0.3$)

Figure 4 Heatmap of global losses under Scenario 2, in absolute values and in percentages: ($\delta = 0.3, \alpha = 0.3$)

β. Similar to Scenario 3, the recovery is heterogeneous with countries like India and Pakistan improving only modestly with the increase in β while others like China and Korea see substantial improvements. scenario 2? Notably, China's output losses decline sharply, from approximately 63,000 million USD to near zero as β increases from 0.05 to 0.1, underscoring the uneven distribution of resilience across economies.

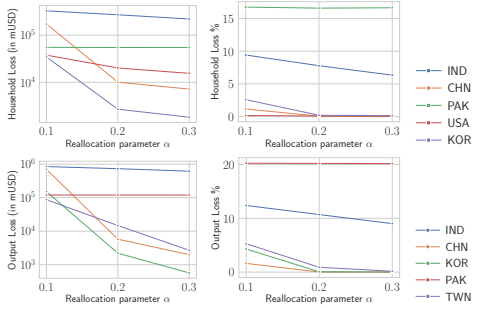
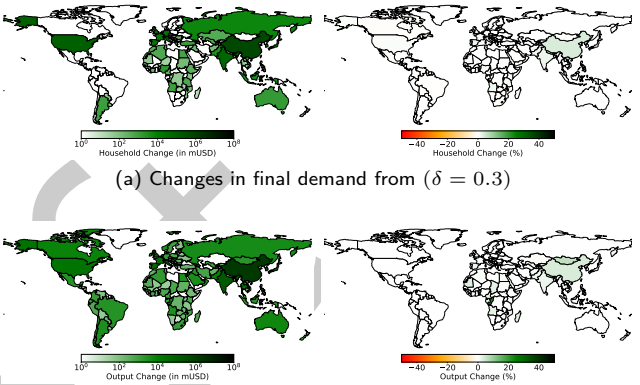


Figure 5 Scenario 2: Top 5 countries (excl. GULF) by loss vs. capacity increase α in TOP-GAS out-flows with fixed $\delta = 0.3$



(a) Changes in final demand from ($\delta = 0.3$)

(b) Output changes from ($\delta = 0.3$)

Figure 6 Heatmap of global losses under Scenario 3, in absolute values and in percentages: ($\delta = 0.3, \alpha = 0.05, \beta = 0.05$) where α is applied globally and β is on TOP-GAS.

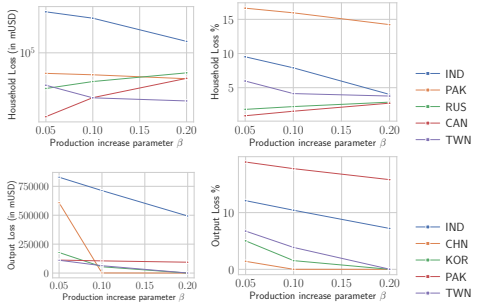


Figure 7 Scenario 3: Top 5 most impacted countries (excl. GULF) by Loss with respect to changes in production (β) in TOP-GAS

Disruption propagation networks

While MRIO models quantify overall impacts, the specific pathways through which disruptions propagate are not immediately captured in the aggregated results. To better understand how shocks propagate in a highly interconnected system, it is useful to study disruption propagation networks within the global input–output network. We define a disruption propagation network as a directed graph consisting of edges (u, v) over which the flow after disruption reduces over a threshold. These networks can help identify critical bottlenecks and amplification nodes where the shock either intensifies or can be effectively mitigated through policy interventions. Below we

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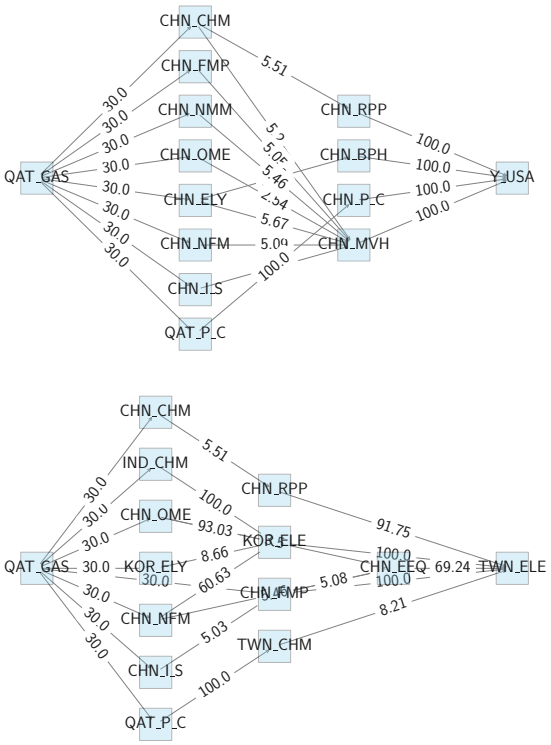


Figure 8 Top 10 disruption propagation paths by aggregate loss (max length = 3, loss threshold =100) under Scenario 1 ($\delta = 0.3$). Edge-labels are the loss in percentages.

study two such subgraphs. One that shows the impact on US final demand and the other on Taiwan's electronics sector.

Figure 8 shows the indirect effects of a 30% disruption in Qatar's gas sector into chemicals, machinery, electronics and final demand across economies. The immediate impact falls on China's chemical sector (CHN-CHM) and Qatar's petroleum and coal products sector (QAT-P-C). The initial disruption is amplified as it propagates through the network. Despite originating from a 30% disruption, this leads to a 100% disruption in the flow from China's Rubber and Plastic Products manufacturing sector (CHN-RPP) to final demand of the US (y_USA) in the first subgraph, and 91.75% from CHN-RPP to Taiwan's electronics sector (TWN-ELE) in the second subgraph. Naphtha, a co-product of LNG production, feeds into petrochemicals and helium into electronic equipment (CHN-EEQ), with losses rippling further to Taiwan's electronics sector (TWN-ELE). Gas supply constraints also propagate through Taiwan's electricity sector (TWN-ELY), disrupting electronics manufacturing and affecting Korea's electronics sector (KOR-ELE). Korea's electronics sector further suffers losses through China's machinery sector (CHN-OME) at 93.03%, reflecting broader network dependence on gas-intensive intermediate production. These results illustrate that a single supply disruption propagates through multiple channels such as feedstocks, energy supply, and intermediate manufacturing sectors among others.

add note about Figure 15

Discussion

Our analyses show that the disruption to the Qatar gas sector can have widespread cascading effects across sectors and regions through global supply chains. The largest impacts are on the large Asian economies of India, Pakistan, China and Korea. These followed by Italy and the US, even though the US doesn't have a direct connection to Qatar, and US is one of the biggest gas producers. Output losses display a similar geographic pattern, with high impact on Taiwan. We observe that reallocation of flows leads to significant benefits to final demands, with China, US, Korea seeing the largest improvements. Finally, production expansion by the top gas producers, and flow reallocation substantially mitigates losses, particularly in large economies such as the US, China, and India. losses in India are not completely reduced even after production increase in Scenario 3 We only say they are mitigated. Unbounded reallocations can potentially lead to significant losses in final demands among some gas producers, but that may be unrealistic. Our results highlight that solution space under increased production and reallocation is very complex, and the final outcomes depend on how countries prioritize output losses versus final demands. Our linear programming and the disruption propagation network analysis methods provide a framework to systematically explore potential solutions and tradeoffs, and understand dependencies between sectors.

Aside from LNG, Ras Laffan jointly produces condensate, helium, liquefied petroleum gas, Naphtha and Sulphur, which will lead to export reductions of 24%, 14%, 13%, 6% and 6% respectively (4; 3). These disruptions propagate downstream affecting household energy use and different sectors including transportation, petrochemicals, and semiconductors among others. In Japan and South Korea, where Naphtha imports are heavily sourced from the Middle East, production of ethylene, PVC, and polyethylene, which are inputs to plastics, packaging and construction materials have also been disrupted (17). Taiwan and South Korea import approximately 69% and 64.7% of their helium, a critical input in semiconductor fabrication, from Qatar respectively, making both countries' chip industries directly exposed. Disruptions in these materials need to be considered along with gas disruptions, in order to understand the full economic impact.

We note that there are several caveats in the counterfactual scenarios on reallocation and production increase we study here. Aside from Qatar, the US and Australia are also major LNG producers, but neither can immediately fill the supply gap. Both countries are operating close to capacity with production already committed under long-term contracts (18). Reallocation is further constrained by energy security priorities. LNG exports are not purely market-driven, as governments may intervene to prioritize domestic supply during periods of shortage, consistent with policies such as Australia's Domestic Gas Security Mechanism (7).

We also note that there are challenges in studying disruptions in other materials associated with gas production. For instance, helium supply does not necessarily increase with natural gas production. Helium is only recoverable from gas streams with sufficiently high concentrations and requires dedicated processing infrastructure, which means that only a limited number of gas fields are commercially viable for extraction (9).

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Despite being a major LNG exporter, Australia does not have the infrastructure to capture helium from its gas production (6). These supply side rigidities mean that disruption cannot be resolved through market reallocation alone, and its economic consequences will propagate across global production networks for an extended period. LNG liquefaction projects take an average of 4-5 years between the final investment decision and completion, with no new capacity expected to come online on a sufficient scale to address the disruption in the next 2-3 years (11). While these constraints limit immediate supply-side responses, understanding the potential mitigating effects of partial trade reallocation and capacity expansion remains critical for policy. Thus, we consider scenarios to quantify the global economic impact of disruption and evaluate the extent to which these adaptive reponses could reduce its consequences. Add a paragraph on the limitations here.

Methodology

We examine how a disruption in Qatar's gas sector affects the global economy. We conceptualize shocks and disruptions as unfolding in three phases: (1) no adaptation, (2) adaptation through increased trade flow capacity, and (3) adaptation through both trade flow and production capacity expansion.

In all scenarios, the initial shock is implemented by reducing flows from the Qatar gas sector (`QAT-GAS`) to all downstream firms and final demand. The magnitude of this disruption is defined as a fraction of baseline flows, denoted by δ. In Scenario 2, we introduce adaptive responses through trade reallocation. Specifically, we increase the flow capacity from a group of major gas-producing countries, denoted as `TOP-GAS`, by α% This set includes the six largest producers outside Qatar: the United States, Russia, Norway, Canada, Australia, and Algeria.

In Scenario 3, we further allow for supply-side adjustment by increasing production capacity within the `TOP-GAS` group. This is modeled as a β% increase in the availability of primary inputs to these sectors. However, since production expansion may be constrained by upstream and downstream network linkages, we simultaneously relax global trade flow capacities by setting $\alpha = \beta$%. This ensures that the additional production can be effectively distributed across the network.

Together, these scenarios enable a structured analysis of how global supply chains respond to energy disruptions, distinguishing between immediate impacts and the role of adaptive mechanisms in mitigating them.

Input-Output Network

We employ a multiregional input-output (MRIO) model (14). The MRIO table is obtained from the newly released GTAP 12 database for the year 2023, which covers 163 countries and 65 sectors (1). We consider an economy consisting of n sectors, denoted by $\mathcal{S}$ and r regions or countries, denoted by $\mathcal{R}$. Each sector $S \in \mathcal{S}$ produces a unique good, e.g., steel, gas etc. A *firm* is a sector in a country, identified by a tuple (country, sector), e.g. `USA-GAS`, `CHN-CHM`; the set of firms is denoted by $\mathcal{F}$. A firm produces its goods using intermediate inputs from other firms and primary inputs such as labor, capital etc. The gross output x_i of a firm i is the sum of its intermediate inputs and value added. We model the flow of goods between firms and end-users as a directed weighted network. There are two types of nodes: firms and final demand (consumed by households), and two types of edges: firm-to-firm, denoted by E_I, and firm-to-final demand, denoted by E_H. Each edge carries a flow of goods whose maximum value cannot exceed the stated edge capacity. Each firm sends its goods along this network to other firms (intermediate flows) and final demand regions (final consumption flows). The flow value from firm i to j is denoted by $z_{i,j}$, flow from firm i to final demand region r is denoted by $h_{i,r}$. In the Leontief framework, the core assumption is that each firm operates on fixed production recipes based on known input coefficients. In our model, products of the same sector sourced from different countries are substitutable. Thus, an input coefficient $A_{S,i}$ is computed as the amount of input required from sector S to produce one unit of output in firm i, i.e., $A_{S,i} = \sum_{j \in S} z_{j,i}/x_i$. The output of a firm is distributed to other firms and final demand as $x_i = \sum_{(i,j)\in E_I} z_{i,j} + \sum_{(i,r)\in E_H} h_{i,r}$.

LP formulation

We model a disruption as a change in flow-capacities for a given subset of edges. Given a disruption, our goal is to find an equilibrium, i.e., the allocation of flows $\{\mathbf{z}, \mathbf{h}\}$ and firm outputs $\mathbf{x}$ on the IO network such that the solution is optimal in some sense. We adopt an optimization approach where we want to find the flow allocations and firm outputs which maximize the total output and total final demand flows. We formulate this bi-objective problem as a weighted combination of the two with parameters λ_1 and λ_2 controlling the importance given to intermediate demand flows versus final demand flows. In this study we keep $\lambda_1 = \lambda_2$ i.e. give equal weight to both intermediate and final demand. Due to the linearity of the Leontief framework, this problem can be formulated as a linear program as shown in Table 3. The Leontief constraints ensures that the solution flows adhere to the input coefficients for every sectoral input flow and firm pair. The conservation of flow constraints ensure that everything that is produced by any firm is distributed to other firms and final demand regions along its outgoing edges, i.e., there is no wasted production.

Scaling to large networks: We modify the standard Leontief equilibrium condition to allow slack in the form of wasted inputs, i.e., input flows that cannot be utilized by the current production level, given by $\epsilon_{S,i} = \sum_{j \in S} z_{j,i} - A_{s,i} x_i$. This is done to speed up the computation of the solution. These wasted input flows are limited to a maximum value, W_{max}, and are penalized in the objective to minimize them. We observe a tradeoff in speed and accuracy when we increase W_{max} from 100 to 0 mUSD, the computation time reduces by up to 20 times. Do you mean 0 to 100 mUSD? How does accuracy increase if you allow more waste?

Dataset

The multiregional input-output (MRIO) table is obtained from the newly released GTAP 12 database, which covers 163 countries and 65 sectors with 2023 as the reference year (1). Each country has a representative firm for each sector, and a representative household for final demand. The number of firms in our model is 10,595. We treat the flows in this table (also referred to as the original values) as the edge capacities, except where it has been disrupted or is adaptive in our scenarios.

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Sets	$\mathcal{F}$	Set of all firms
	$S \subseteq \mathcal{F}$	Set of firms with same product called *sector*
	$\mathcal{F}_{GAS}$	Set of TOP-GAS firms
	$\mathcal{S}$	Set of all sectors
	$\mathcal{R}$	Set of all final demand regions
	$E_I \subseteq \mathcal{F} \times \mathcal{F}$	Set of all firm-to-firm trade edges
	$E_H \subseteq \mathcal{F} \times \mathcal{R}$	Set of all firm-to-region edges
	E_δ	Set of disrupted edges
	E_α	Set of edges with capacity increases
	$N_I^+(i), N_I^-(i), N_H^+(i), N_H^-(i)$	Set of outgoing (or incoming) neighbors of node i across edges in E_I (or E_H)
Variables	$x_i \ \forall i \in \mathcal{F}$	Output of each firm i
	$z_{i,j} \ \forall (i,j) \in E_I$	Flow from firm i to firm j
	$h_{i,r} \ \forall (i,r) \in E_H$	Flow from firm i to final demand region r
	$\epsilon_{S,i} \ \forall (S,i) \in \mathcal{S} \times \mathcal{F}$	Wasted input from sector S to firm i
Constants	$A_{S,i} \ \forall S \in \mathcal{S}, i \in \mathcal{F}$	Input coefficient of firm i from sector S
	$B_i \ \forall i \in \mathcal{F}$	Primary input coefficient of firm i
	$P_i^*, Z_{i,j}^*, H_{i,r}^*, X_i^*$	MRIO baseline values for primary inputs, intermediate flows, final demand flows, firm outputs resp.
	δ	Flow capacity disruption
	α	Flow capacity increase
	β	Primary input capacity increase
	W_{max}	Max allowable wasted flow for a (sector, flow) pair

Table 2. Table of symbols in the LP 3

Thresholding both inter-firm, final demand flows, input coefficients.

Computational speed-ups

Thresholds, slack on wasted input.

Model assumptions

Fixed production recipes, input coefficients don't vary with time, Multiple solutions possible,

Datasets

Experimental setup

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maximize	$\underbrace{\sum_{i\in\mathcal{F}} x_i}_{\text{Production incentive}} + \underbrace{\lambda_1 \sum_{(j,k)\in E_I} z_{j,k}}_{\text{Intermediate flow incentive}} + \underbrace{\lambda_2 \sum_{(l,r)\in E_H} h_{l,r}}_{\text{Final demand flow incentive}}$ $\underbrace{-\lambda_3 \sum_{(S,i)\in\mathcal{S}\times\mathcal{F}} \epsilon_{S,i}}_{\text{Penalty for wasted inputs}}$	
subject to,		
Leontief constraints:	$A_{S,i}x_i + \epsilon_{S,i} = \sum_{j\in S\cap N_I^-(i)} z_{j,i}$	$\forall S\in\mathcal{S}, i\in\mathcal{F}$
	$B_i x_i \le P_i^*$	$\forall i\in\mathcal{F}\setminus\mathcal{F}_{GAS}$
	$B_i x_i \le (1+\beta)P_i^*$	$\forall i\in\mathcal{F}_{GAS}$
Conservation of flow:	$\sum_{j\in N_I^+(i)} z_{i,j} + \sum_{r\in N_H^+(i)} h_{i,r} = x_i$	$\forall i\in\mathcal{F}$
Flow capacity bounds:	$z_{i,j} \le Z_{i,j}^*$	$\forall (i,j)\in E_I\setminus(E_\delta\cup E_\alpha)$
	$h_{i,j} \le H_{i,j}^*$	$\forall (i,j)\in E_H\setminus(E_\delta\cup E_\alpha)$
	$\epsilon_{S,i} \le W_{max}$	
Disruption & adaptation:	$z_{i,j} \le (1-\delta)Z_{i,j}^*$	$\forall (i,j)\in E_I\cap N_i^+$
	$h_{i,r} \le (1-\delta)H_{i,r}^*$	$\forall (i,r)\in E_H\cap E_\delta$
	$(1-\alpha)Z_{i,j}^\delta \le z_{i,j} \le (1+\alpha)Z_{i,j}^*$	$(i,j)\in E_I\cap E_\alpha$
	$(1-\alpha)H_{i,r}^\delta \le h_{i,r} \le (1+\alpha)H_{i,r}^*$	$(i,r)\in E_H\cap E_\alpha$

Table 3. The IO LP for the study of trade disruptions and adaptations.

$\epsilon_{s,i} \le$ U.B.	Penalty Scale (λ_3)	Total Production	Wasted Input Flows ($\sum \epsilon_{s,i}$)	Time to solve
0	No slack allowed	187,105,695	0 (0.00%)	8 hrs
N.B.	0	187,792,119	2883500 (1.54%)	10 min
N.B.	1e-6	187,792,119	635486.47 (0.34%)	10 min
N.B.	1e-3	187,792,105	634412.29 (0.34%)	10 min
N.B.	1	187,792,100	629593.56 (0.34%)	20 min
100	1e-3	187,250,102	147,036 (0.08%)	20 min
10	1e-3	187151721	43416 (0.02%)	1.50 hr

Table 4. To minimize "wasted" inputs, we add a penalty term $-\lambda_3 \sum \epsilon_{s,i}$ to the objective. These are the solutions to different penalty scales, i.e. λ_3. All figures are in mUSD.

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Technical Appendix

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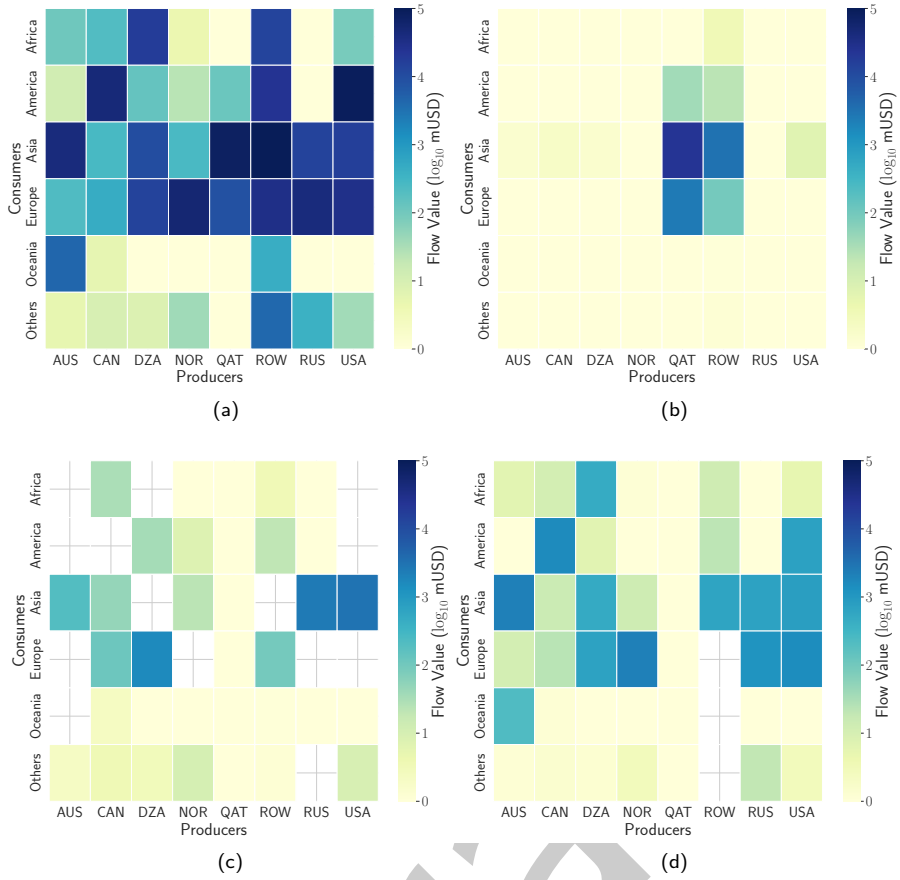


Figure 9 Producer to Consumer Regions (grouped by Continent) (a) Flows at $(\delta = 0)$. (b) Decline in flows at $(\delta = 0.3)$. (c) Change in flows at $(\delta = 0.3, \alpha = 0.3)$ w.r.t. $(\delta = 0.3)$. (d) Change in flows at $(\delta = 0.3, \alpha = 0.05, \beta = 0.05)$ w.r.t $(\delta = 0.3)$.

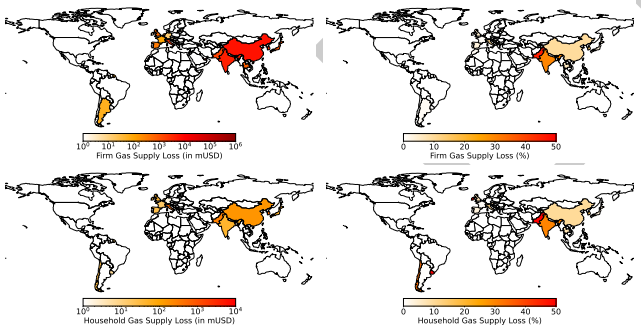


Figure 10 Scenario 1: Loss in Gas Flows to Firms and households under disruption $\delta = 0.5$



Figure 12 Scenario 3: Changes in Gas Flows to Firms and Households in $(\delta = 0.3, \alpha = 0.05, \beta = 0.05)$ from $(\delta = 0.3)$.

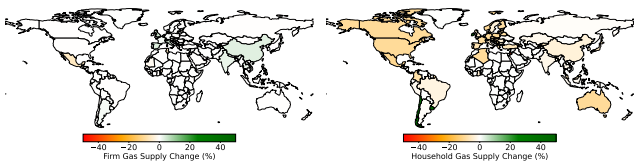
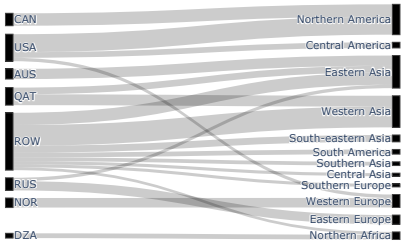
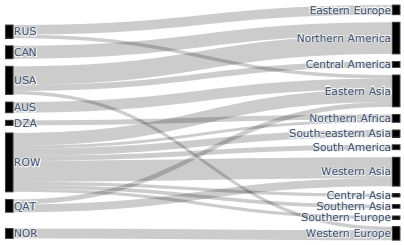


Figure 11 Scenario 2: Changes in Gas Flows to Firms and Households in $(\delta = 0.5, \alpha = 0.1)$ from $(\delta = 0.5)$.



(a)



(b)

Figure 13 Gas Flows from Producers to Consumer Regions at (a) $\delta = 0$, (b) $\delta = 0.3$. **RM**: Sankey diagrams not useful for comparison between flows in different scenarios.

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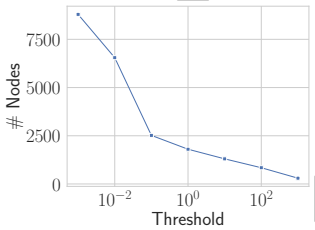


Figure 14 Cascade size with varying loss threshold

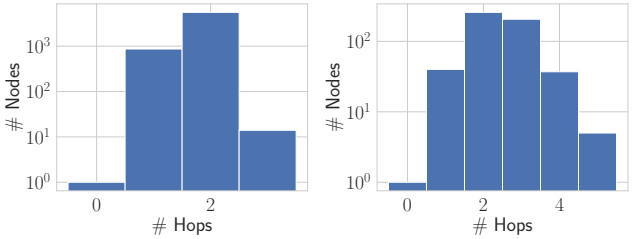


Figure 15 a. loss threshold = 0.01, b. loss threshold =100